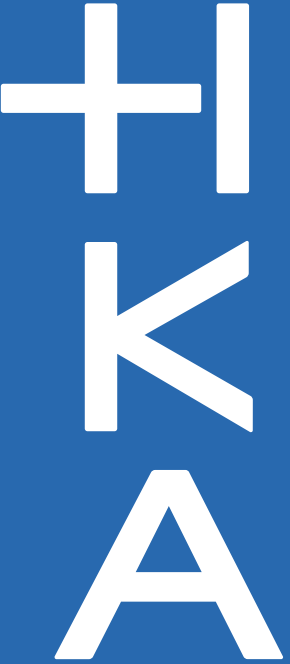
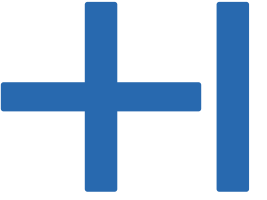


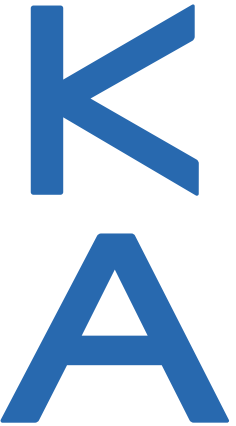
How to use the System





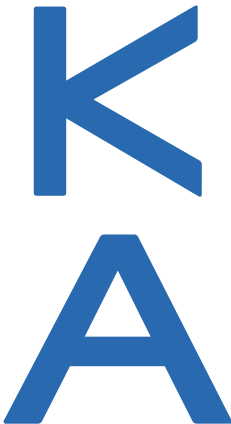
Start the program in the Terminal using:

```
● (base) PS C:\Master\F-E_Projekt1> conda activate F+E  
○ (F+E) PS C:\Master\F-E_Projekt1> ./start_all.ps1
```



Important: You must be in the project folder

This is the interface



DataBridge Explorer

FileView

DataBridge Explorer

Knowledge netProjection

Job from .h5ad ...

Nothing selected yet.
Check cohorts and choose attributes on the right.

ArchitectureText output

Auswahl → JSON
frontend/mediator_client.py

Mediator
FastAPI · POST /selection/*

Wrapper → Datenquelle
wrappers/qdc · GDC-API

GDC-JSON → RDF
app/semantic/mapping.py

graph-db (Fuseki)
Apache Jena · Default-Graph

Wissensnetz
wissensnetz · SPARQL

Noch kein Aufruf. Die Kette zeigt, was beim Abschicken der Reihe nach passiert.

PreviewGenerateSave as .h5ad

EnglishBereit. Mediator; http://localhost:8000 · 32 Kohorten geladen.

Store: 18 Faele · 1 Kohorte

Cancer
No selection

Var
Genexpression

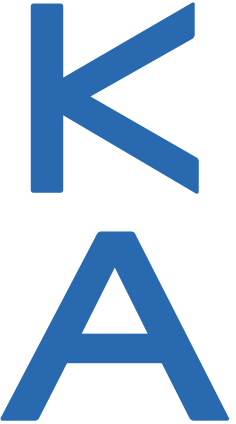
Obj
No selection

Data source
No selection

Samples
20

Pablo Scherer | Marcel Luis Thiel | Julian Lanfermann

You can set the languages here



DataBridge Explorer

File View

DataBridge Explorer

Knowledge net

Projection

Structure and counts, no measurement data

Job from .h5ad ...

Nothing selected yet.
Check cohorts and choose attributes on the right.

Architecture

Text output

What is happening in the background

Auswahl → JSON
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wissensnetz · SPARQL

Noch kein Aufruf. Die Kette zeigt, was beim Abschicken der Reihe nach passiert.

Preview

Generate

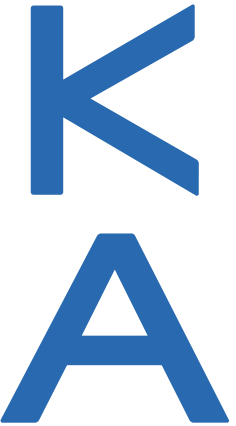
Save as .h5ad

English

Bereit. Mediator: http://localhost:8000 · 32 Kohorten geladen.

Store: 18 Faele · 1 Kohorte

You can set the Cancer here



Cancer

No selection

Var

Genexpression

Obj

No selection

Data source

No selection

Samples

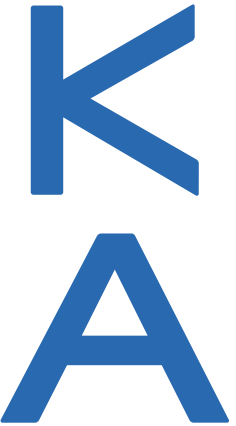
20

Cancer

No selection

Search code, e.g. BRCA ...

☐	Adrenocortical Carcinoma	ACC
☐	Cholangiocarcinoma	CHOL
☐	Bladder Urothelial Carcino...	BLCA
☐	Breast Invasive Carcinoma	BRCA
☐	Cervical Squamous Cell C...	CESC
☐	Colon Adenocarcinoma	COAD
☐	Uterine Corpus Endometr...	UCEC
☐	Esophageal Carcinoma	ESCA
☐	Glioblastoma Multiforme	GBM



Cancer

No selection

Var

Genexpression

Obj

No selection

Data source

No selection

Samples

20



Cancer

No selection

Var

Genexpression
DNA-Methylierung (noch nicht angebunden)
Mutationen (noch nicht angebunden)
No selection

Data source

No selection

Samples

20

Cancer

No selection

Var

Genexpression

Obj

No selection

Data source

No selection

Samples

20



Cancer

No selection

Var

Genexpression

Obj

No selection

Search attribute or node, e.g. stage ...

Demographic

☐ sex_at_birth

☐ race

☐ ethnicity

☐ vital_status

Diagnosis

☐ primary_diagnosis

☐ age_at_diagnosis

☐ morphology

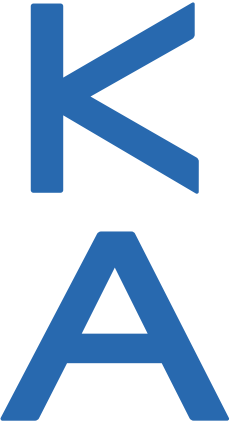
☐ site_of_resection_or_biopsy

☐ tumor_stage

☐ has_metastasis

Sample

☐ sample_type



You can set the Data source here
But cBioPortal and GEO not available



Cancer

No selection

Var

Genexpression

Obj

No selection

Data source

No selection

Samples

20



Cancer

No selection

Var

Genexpression

Obj

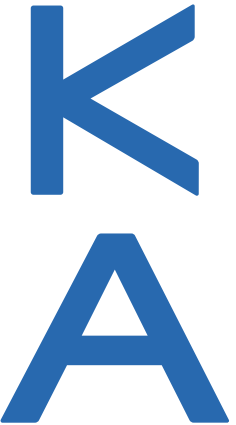
No selection

Data source

No selection

- ☐ GDC / TCGA
- ☐ cBioPortal
- ☐ GEO
- ☐ ENA nicht angebunden

After you've set the variables, you can click “Preview” to see if the system has found anything.



DataBridge Explorer

File View

DataBridge Explorer

Knowledge net Projection

Structure and counts, no measurement data

Job from .h5ad ...

Nothing selected yet.
Check cohorts and choose attributes on the right.

Architecture Text output

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Preview Generate Save as .h5ad

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No selection

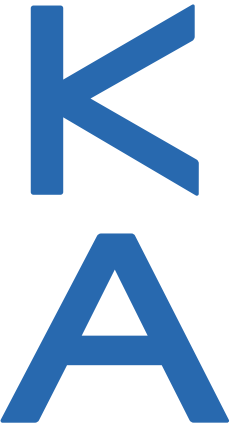
Var
Genexpression

Obj
No selection

Data source
No selection

Samples
20

After setting the variables, you can click “Generate,” and the memory module next to it will become active, allowing you to save the “Anndata.”



DataBridge Explorer

File View

DataBridge Explorer

Knowledge net Projection

Job from .h5ad ...

Nothing selected yet.
Check cohorts and choose attributes on the right.

Architecture Text output

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Auswahl → JSON
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Structure and counts, no measurement data

Cancer
No selection

Var
Genexpression

Obj
No selection

Data source
No selection

Samples
20

You can also view the data using a projection, but only if you've saved the “Anndata” beforehand.

